FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – XI
PAPER –2
TEST DATE: 28-05-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

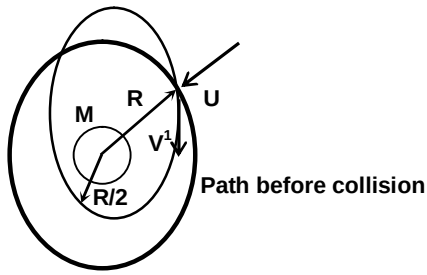
1. A

Sol. We know that Activity $A = N\lambda$ and $T_{1/2} = \frac{\ln 2}{(A/N)} = \frac{N\ln 2}{A}$

$N_p = \frac{It}{q}$ and $N = \frac{N_p}{1000}$

2. C

Sol. Situation before and after collision is shown in figure.



Before collision the speed of satellite is $U_0 = \sqrt{\frac{GM}{R}}$

Consider the meteorite collide with speed u and after collision the combined mass 11m moves at an angle θ with the orbit.

Apply LCM $mu = 11mV^1\sin\theta$ -----(1)

and $10mU_0 = 11mV^1\cos\theta$ -------(2)

Apply LCAM, $11m(V^1\cos\theta)R = 11mV_p(R/2)$ ----(3) here V_p is speed at perigee.

Apply LCE, $-\frac{GM11m}{R} + \frac{1}{2}11mV^{1^2} = -\frac{GM11m}{R/2} + \frac{1}{2}11mV_p^2$ ----(4)

Solving $U = \sqrt{\frac{58GM}{R}}$

3. C

Sol. Consider a small element dx at a distance x from the long wire.

$$\tau = \int_L^{2L} F.(2L-x) \Rightarrow \tau = \int_L^{2L}\left(\frac{\mu_0 i}{2\pi x}\right)idx.(2L-x) = \frac{\mu_0 i^2}{2\pi^2}\left[2L\log 2 - L\right]$$

And M.I.$= \frac{mL^2}{3}$, use $\tau = I\alpha$.

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4. A

Sol. Flux through an rectangular element of width dx at a distance x from P is $\phi = \int \vec{E} \cdot d\vec{s}$

$ds = ldx$ and $E = \frac{\lambda}{2\pi\varepsilon_0 d\sec\theta}$, $\phi = \int_{-b/2}^{b/2} \frac{\lambda}{2\pi\varepsilon_0 d\sec\theta}\cos\theta ldx = \frac{\lambda\ell}{\pi\varepsilon_o}\tan^{-1}\left(\frac{b}{2d}\right)$

5. A, C, D

Sol. $U_i = \frac{Qq}{4\pi\varepsilon_0 r} + \frac{Q^2}{8\pi\varepsilon_0 r}$ and $U_f = \frac{Qq}{4\pi\varepsilon_0 R} + \frac{Q^2}{8\pi\varepsilon_0 R} \Rightarrow dU = U_f - U_i$ and $W_{el} = -dU$

6. A, D

Sol. The magnitude of induced electric field $\int E \cdot dl = \frac{d\phi}{dt} \Rightarrow E = B_0 Rt$

The tangential force on the ring $F = EQ = B_0 QRt$

Torque due to friction is μmgR. The ring starts rotating when the torque due to F is greater than torque due to friction. $\tau_F \geq \tau_f$

$FR = \mu mgR \Rightarrow \mu = \frac{2B_o RQ}{mg}$

After 2s net torque $\tau = \tau_F - \tau_f = B_0 QR^2(t-2) = I\alpha = \frac{d\omega}{dt}$. At t=4s, $\omega = \frac{2B_0 Q}{m}$

Now magnetic field is switched off, Only retarding torque due to friction is present.

$\alpha = \tau / I = \frac{\mu g}{R}$, use $\omega_2^2 - \omega_1^2 = 2\alpha\theta$ and $\theta = \frac{B_0 Q}{m}$

7. A, C

Sol. $N = N_0 e^{-\lambda t}$, $\frac{dN}{dt} = -N_0\lambda e^{-\lambda t}$, $A = \log\left|\frac{dN}{dt}\right| = \log N_0\lambda - \lambda t = -\lambda t + \log N_0\lambda$

Slope of the straight line is $\lambda = (A_0/2)/t_0 = \frac{A_0}{2t_0}$

At t=0, $A_0 = \log\left|\frac{dN}{dt}\right| = \log(N_0\lambda) \Rightarrow N_0 = \frac{1}{\lambda}e^{A_0} = \frac{2t_0}{A_0}e^{A_0}$

After half-life $N = \frac{t_0}{A_0}e^{A_0}$

8. A, C

Sol. Apply law of refraction at spherical surface

$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

9. A, C, D

Sol. In Bohr model $r\alpha\frac{n^2}{Z}$, $v\alpha\frac{Z}{n}$ and $\omega = v/r$

M = iA and $B = \frac{\mu_0 i}{2r}$

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10. A, D

Sol. Let power ratings of two heaters P_1 and P_2, $H = P_1t_1 = P_2t_2$

Now, when they are connected in series, $H = \dfrac{V^2}{\left(\dfrac{V^2}{P_1}+\dfrac{V^2}{P_2}\right)}t = \dfrac{P_1P_2}{P_1+P_2}t \quad \Rightarrow t = t_1 + t_2$

When they are connected in parallel $H = \dfrac{V^2(R_1+R_2)}{R_1R_2}t \Rightarrow t = \dfrac{t_1t_2}{t_1+t_2}$

Section – B

11. 3

Sol. Current is in clockwise, $i = \dfrac{B\ell(V_1^1 - V_2^1)}{R}$

Now, $Bi\ell = -m\dfrac{dV_1^1}{dt}$ and $Bi\ell = m\dfrac{dV_2^1}{dt}$

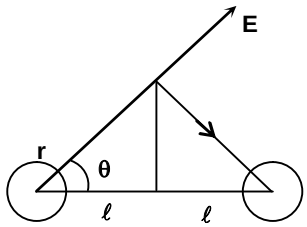
$$Bi\ell\left[\frac{1}{m}+\frac{1}{m}\right] = -\frac{d(V_1^1 - V_2^1)}{dt} \Rightarrow \frac{B\ell B\ell(V_1^1 - V_2^1)}{R}\left[\frac{1}{m}+\frac{1}{m}\right] = -\frac{d(V_1^1 - V_2^1)}{dt}$$

$$\frac{d(V_1^1 - V_2^1)}{(V_1^1 - V_2^1)} = -\frac{2B^2\ell^2}{mR}dt \qquad (V_1^1 - V_2^1) = (V_1 - V_2)e^{-\left(\frac{2B^2\ell^2t}{mR}\right)}$$

$$i(t) = \frac{B\ell(V_1 - V_2)e^{-\left(\frac{YB^2\ell^2t}{mR}\right)}}{XR} \Rightarrow X + Y = 3.$$

12. 2

Sol. Since balls are far apart, we may take charge is uniformly distributed on the balls



$$E = \frac{2q\cos\theta}{4\pi\in_0\in r^2} = j\rho$$

$$\Rightarrow j = \frac{ql}{2\pi\in_0\in \rho r^3}$$

Total current across the plane $i = \int_0^\infty j2\pi x dx$

$$\Rightarrow i = \frac{q}{\in\in_0\rho} \qquad \therefore V[\rho d] = \frac{q}{2\pi\in_0 a} \qquad \therefore R = \frac{V}{i} = \frac{\rho}{2\pi a}$$

13. 7

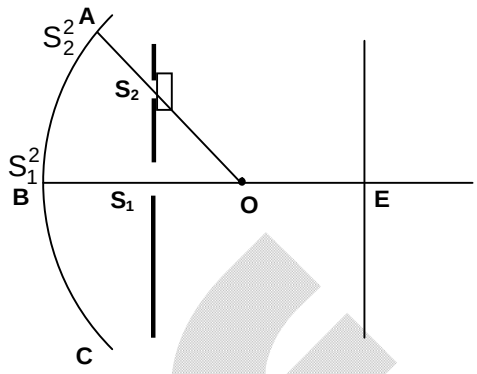
Sol. By covering aperture, focal length does not change.

$I \alpha$ area of aperture $\dfrac{I^1}{I} = \dfrac{A^1}{A} = \dfrac{3}{4}$

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14. 8

Sol. Let S_1^1 and S_2^1 are the points on the wave front where perpendiculars can be drawn from S_1 and S_2. For central fringe formed at E path difference should be zero.



$$S_2^1S_2 + (\mu - 1)t + S_2E = S_1^1S_1 + S_1E$$

$$(OS_2^1 - OS_2) + (\mu - 1)t + \sqrt{D^2 + d^2} = (OS_1^1 - OS_1) + D$$

$$(\mu - 1)t = (OS_2 - OS_1) - (\sqrt{D^2 + d^2} - D)$$

using binomial approximation $t = \frac{31\lambda}{8}$

15. 2

Sol. At an instant let i be the current through the battery and i_1 be through inductor and i_2 through resistor. $i = i_1 + i_2$. Apply KVL for the two loops

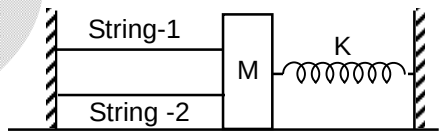
$$-L\frac{di_1}{dt} - i_1R - iR + \varepsilon = 0 \text{ ---(1)} \quad \text{and} \quad i_2R - iR + \varepsilon = 0 \text{ solving } i_1 = 2i - \frac{\varepsilon}{R}, \text{ sub this in (1)}$$

$$-L\frac{d}{dt}\left(2i - \frac{\varepsilon}{R}\right) - \left(2i - \frac{\varepsilon}{R}\right)R - iR + \varepsilon = 0 \Rightarrow \int_{E/2R}^{i} \frac{di}{E - \frac{3}{2}iR} = \int_0^t \frac{dt}{L}$$

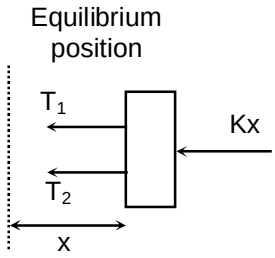
$$i = \frac{E}{2R} + \frac{E}{6R}\left(1 - e^{-3Rt/2L}\right)$$

16. 9

Sol. In this question the most important point to keep in mind is that there is no tension developed in string if string gets sag. In present situation if we displace the block by x towards right then both the strings get taut & spring gets compressed as a result all three components (two strings & a spring) contribute in restoring force, but when the block comes towards left of its equilibrium position then string got stretched and only the spring force is helping in bringing the block to its equilibrium position. It means equation of motion for the displacement of block on two sides of its equilibrium position is different.



For displacement towards right by x –



$$\frac{T_1 \,/\, A}{x/\ell} = Y \Rightarrow T_1 = \frac{YAx}{\ell}$$

$$\frac{T_2 \,/\, A/2}{x/\ell} = 2Y \Rightarrow T_2 = \frac{YAx}{\ell}$$

Restoring force,

$$F = T_1 + T_2 + Kx = \left(\frac{2yA}{\ell} + K\right)x$$

$$\Rightarrow a = \frac{1}{m}\left(\frac{2YA}{\ell} + K\right)x$$

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$$\omega_1 = \sqrt{\frac{1}{m}\left(\frac{2YA}{\ell} + K\right)}$$

$$\Rightarrow T_1 = 2\pi\sqrt{\frac{M\ell}{2AY + K\ell}} = \pi\sqrt{\frac{M}{K}}$$

For displacement towards left by x- the situation is similar to a spring pendulum with time period,

$$T_2 = 2\pi\sqrt{\frac{M}{K}}$$

Required time period of oscillation is,

$$T = \frac{T_1}{2} + \frac{T_2}{2} = \pi\left[\sqrt{\frac{M\ell}{2AY + K\ell}} + \sqrt{\frac{M}{K}}\right] = \frac{3\pi}{2}\sqrt{\frac{M}{K}}$$

17. 3

Sol. $$f_D = \left[\frac{330 + V_w}{330 + V_w - V_s \sin 30^o}\right] f$$

$$\% \text{ increase} = \left(\frac{f_D - f}{f}\right) \times 100$$

18. 1

Sol. $$2mg\ell\sin\theta = KE_A - KE_B$$

$$KE_B\Big|_{min} = \frac{1}{2}mu_B^2 = \frac{mg\ell\sin\theta}{2}$$

$$\frac{KE_A}{KE_B} = 5$$

Ratio is independent of 'θ'

So $$\frac{\eta_1}{\eta_2} = 1$$

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Chemistry

PART – II

Section – A

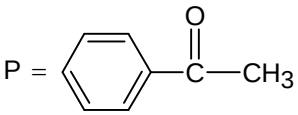
19. D
Sol. Hydroboration - Reduction

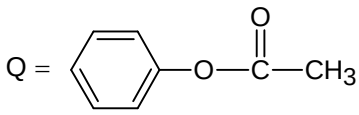
20. D
Sol. $Na_2S + Na_2\left[Fe(CN)_5NO\right] \longrightarrow Na_4\left[Fe(CN)_5NOS\right]$

21. A
Sol. Neoprene is a polymer of chloroprene.

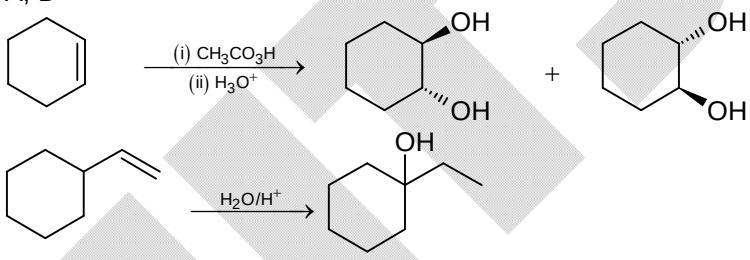
22. A
Sol. Dipole moment of NH_3 is greater than that of NF_3.

23. C, D
Sol.





24. A, D
Sol.



(Major product)

25. A, C, D
Sol. S^{2-} ions forms CCP and Zn^{2+} ions are present in alternate tetrahedral voids in Zinc blende (ZnS)

26. A, B
Sol. When inert gas is added at constant pressure the equilibrium position shift towards product side when $\Delta_n(g) > 0$.

27. A, B, C
Sol. $\log k = \log A - \dfrac{Ea}{2.303RT}$

$\dfrac{Ea}{2.303RT} = \dfrac{3\times10^3}{T}$

$Ea = 2.303R \times 3 \times 10^3 = 13.818$ kcal

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$$Ea = 2.303 \times \frac{2}{1000} \times 3 \times 10^3$$

$$k = \frac{0.693}{t_{1/2}} = \frac{0.693}{30 \times 60}$$

$$k = 3.85 \times 10^{-4} \text{ sec}^{-1}$$

28. A, B, D

Sol. ClO_2 does not dimerise on cooling.

Section – B

29. 8

Sol. Meq. of $KMnO_4$ = Meq. of H_2O_2

$$320 \times \frac{1}{10} \times 5 = 112 \times N$$

$$N = \frac{32 \times 5}{112}$$

Volume strength $N \times 5.6$

$$= \frac{32 \times 5}{112} \times 5.6 = 8$$

30. 5

Sol. $PCl_3 + 3H_2O \longrightarrow H_3PO_3 + 3HCl$

Number of moles required to neutralize

1 mole of $H_3PO_3 = 2$

Number of moles NaOH required to neutralize 3 moles of HCl = 3

Total mole of NaOH required = 2 + 3 = 5

31. 8

Sol. $KE = \frac{3}{2}nRT$

$$\frac{KE_{CH_4}}{KE_{SO_2}} = \frac{n_{CH_4}}{n_{SO_2}} \times \frac{T_{CH_4}}{T_{SO_2}}$$

$$= \frac{4}{1} \times \frac{800}{400} = 8$$

32. 6

Sol. $\Delta T_f = i \times K_f \times m$

$$0.744 = i \times 1.86 \times \frac{2.73}{273} \times \frac{1000}{100}$$

$i = 4$

$\therefore$ Formula of the compound is $\left[Co(H_2O)_6\right]Cl_3$ number of H_2O molecules in coordination sphere is 6.

33. 3

Sol. $6NaOH + 3Cl_2 \longrightarrow 5NaCl + NaClO_3 + 3H_2O$

(Hot and conc.)

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34. 4

Sol. $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

$P_{NH_3} = 2 \times P_{CO_2}$

$P_{Total} = 2P + P = 3P = 3$ bar

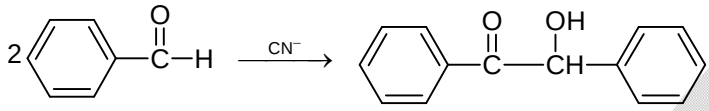
$\therefore P_{NH_3} = 2$ bar and $P_{CO_2} = 1$ bar

$K_P \left(P_{NH_3}\right)^2 \times \left(P_{CO_2}\right)$

$K_P = (2)^2 \times 1 = 4$

35. 7

Sol. Benzoin condensation



(Benzaldehyde) (Benzoin)
(Major product)

Number of π – bonds in one molecule of Benzoin is 7.

36. 3

Sol. Meq. of $NH_3 = (50 \times 0.2 \times 2) - (50 \times 0.2 \times 1)$

= 10

wt. of $NH_3 = \frac{10 \times 17}{1000} = 0.17$ g

Wt. of $N = \frac{14}{17} \times 0.17 = 0.14$ g

% of $N = \frac{0.14}{0.5} \times 100 = 28$

Molecular mass of compound ($C_8H_{12}N_x$)

= 96 + 12 + 14x = 108 + 14x

$\% N = \frac{14x}{108 + 14x} \times 100$

$28 = \frac{14x}{108 + 14x} \times 100$

x = 3.

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Mathematics

PART – III

Section – A

37. C

Sol. $\int_{x_1}^{x_2}\frac{1}{x}dx=\ln\left(\frac{x_2}{x_1}\right)\Rightarrow\int_{a_1}^{b_1}\frac{1}{x}dx=S_1=\left|\ln\left(\frac{b_1}{a_1}\right)\right|$

$S_2=\int_{a_2}^{b_2}\frac{1}{x}dx=\left|\ln\left(\frac{b_2}{a_2}\right)\right|$ as $\frac{b_1}{a_1}=\frac{b_2}{a_2}\Rightarrow S_1=S_2$

38. B

Sol. Let $f(x)=\frac{ax^2+b}{\sqrt{x^2+1}}=a\sqrt{x^2+1}+\frac{b-a}{\sqrt{x^2+1}}$ where a > 0 and f(0) = b clearly $b\geq 3$

Case-I: $b-2a\geq 0\Rightarrow f(x)=a\sqrt{x^2+1}+\frac{b-a}{\sqrt{x^2+1}}\geq 2\sqrt{a(b-a)}=3$

Equality hold if $a\sqrt{x^2+1}=\frac{b-a}{\sqrt{x^2+1}}\Rightarrow x=\pm\sqrt{\frac{b-2a}{a}}$ and value of a for given b is $a=\frac{b-\sqrt{b^2-9}}{2}$

Case-II: If b – 2a < 0, then let $\sqrt{x^2+1}=t$ where $t\geq 1$

$f(x)=g(t)=at+\frac{b-a}{t}$ is monotonic increasing when $t\geq 1$

So, $\min_{x\in R} f(x)=g(1)=a+b-a=b=3$ when $a>\frac{3}{2}\Rightarrow$ Range of b $\in[3,\infty)$

39. D

Sol. Shown in the table

	1st	2nd	3rd	4th
Case (1)	Y	W	W	Y
Case (2)	W	Y	W	Y
Case (3)	W	W	Y	Y

Probability = P(case (1)) + P(case (2)) + P(case (3))

$=\frac{2}{10}\times\left(\frac{9}{10}\right)^2\times\frac{1}{10}+\frac{8}{10}\times\frac{2}{10}\times\frac{9}{10}\times\frac{1}{10}+\left(\frac{8}{10}\right)^2\times\frac{2}{10}\times\frac{1}{10}=0.0434$

40. A

Sol. $\cos\frac{\alpha}{2}\cdot\cos\frac{\alpha}{4}\cdot\cos\frac{\alpha}{8}.....\cos\left(\frac{\alpha}{2^n}\right)=\frac{1}{2^n}\frac{\sin\alpha}{\sin\left(\frac{\alpha}{2^n}\right)}$

$\Rightarrow\lim_{n\to\infty}\cos\frac{\alpha}{2}\cdot\cos\frac{\alpha}{4}\cdot\cos\frac{\alpha}{8}\cdot\cos\frac{\alpha}{16}.....\cos\left(\frac{\alpha}{2^n}\right)=\left(\lim_{n\to\infty}\frac{\left(\frac{1}{2^n}\right)}{\sin\frac{\alpha}{2^n}}\right)\sin\alpha=\frac{\sin\alpha}{\alpha}$

$\Rightarrow\cos\frac{\alpha}{2}=\sqrt{\frac{1}{2}+\frac{1}{2}\cos\alpha}\Rightarrow\cos\frac{\alpha}{4}=\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}+\frac{1}{2}\cos\frac{\alpha}{2}}}$ similarly all of terms

Now, put $\alpha=\frac{\pi}{2}$

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41. A, D

Sol. $P_n(r) = pP_{n-1}(r+1) + qP_{n+1}(r+1)$. Let $f_n(x) = \sum_{r=-\infty}^{\infty} P_n(r)\cdot x^r = f_{n-1}(x)\cdot\left(Px+\frac{q}{x}\right)$

As $f_n(x) = \left(Px+\frac{q}{x}\right)^n f_0(x) \Rightarrow f_n(x) = \left(Px+\frac{q}{x}\right)^n$ and $F_0(x) = P_0(x) = 1$

$\Rightarrow P_n(r)$ is the coefficient of x^r in $\left(Px+\frac{q}{x}\right)^n$

42. A, D

Sol. $T_{n+1} = T_n(T_n - 1) + 1 = T_n(T_{n-1}\ T_{n-2}\ T_{n-3}\ .....\ T_1) + 1 = 1 + T_1\ T_2\ T_3\ .....\ T_n$ and $\sum_{r=1}^{n}\frac{1}{T_r} = 1 - \frac{1}{T_{n+1}-1}$

$$\left\{\sum_{r=1}^{k+1}\frac{1}{T_r} = 1 - \frac{1}{T_{k+1}-1} + \frac{1}{T_{k+1}} = 1 - \left\{\frac{T_{k+1} - (T_{k+1}-1)}{T_{k+1}(T_{k+1}-1)}\right\} = 1 - \frac{1}{T_{k+1}(T_{k+1}-1)} = 1 - \frac{1}{T_{k+2}-1}\right\}$$

As T_n tend to infinity $\sum_{r=1}^{\infty}\frac{1}{T_r}$ tend to 1

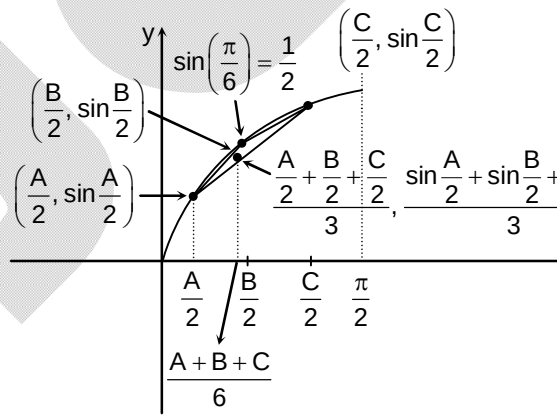
43. A, D

Sol. $0 \le \frac{A}{2}, \frac{B}{2}, \frac{C}{2} \le \frac{\pi}{2}$

$$\frac{\sin\frac{A}{2} + \sin\frac{B}{2} + \sin\frac{C}{2}}{3} \le \sin\left(\frac{A+B+C}{6}\right)$$

$$\sin\frac{A}{2} + \sin\frac{B}{2} + \sin\frac{C}{2} \le \frac{3}{2}$$

Similar for function sine and cosine both are concave in $\left(0, \frac{\pi}{2}\right)$



44. C, D

Sol. $|z| \le 2$

$\Rightarrow (a + \cos\theta)^2 + (2a - \sin\theta)^2 \le 4 \Rightarrow 2a(\cos\theta - 2\sin\theta) \le 3 - 5a^2$

$\Rightarrow \frac{2a}{\sqrt{5}}\sin(\phi - \theta) \le 3 - 5a^2 \Rightarrow |a| \le \frac{1}{\sqrt{5}}$

45. A, C

Sol. Put $x = y = 1 \Rightarrow f^2(1) + f^2(a) = 2f(1)$

$\Rightarrow (f(1) - 1)^2 = 0 \Rightarrow f(1) = 1$. Now, by putting $y = \frac{a}{x}$

$$f(x)f\left(\frac{a}{x}\right) + f\left(\frac{a}{x}\right)f(x) = 2f(a) \Rightarrow f(x)f\left(\frac{a}{x}\right) = 1$$

By $y = 1$

$$f(x)\cdot f(1) + f\left(\frac{a}{x}\right)\cdot f(a) = 2f(x) \Rightarrow f(x) = f\left(\frac{a}{x}\right)$$

$\Rightarrow f^2(x) = 1 \Rightarrow f(x) = 1$ only

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46. A, C

Sol. Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(t^2, 2t)$ from $x - 2y - 1 = 0$ and $y^2 = 4x$, we get $y^2 - 8y - 4 = 0$

$\Rightarrow y_1 + y_2 = 8$, $y_1y_2 = -4$ and $x_1 + x_2 = 2(y_1 + y_2) + 2 = 18$ by $(x - 2y - 1 = 0)$

$x_1x_2 = 4y_1y_2 + 2(y_1 + y_2) + 1 = 1$

Further $\angle ACB = 90^\circ$ so $\overrightarrow{CA} \cdot \overrightarrow{CB} = 0$

$\Rightarrow (t^2 - x_1)(t^2 - x_2) + (2t - y_1)(2t - y_2) = 0$

$\Rightarrow t^4 - 14t^2 - 16t - 3 = 0 \Rightarrow t^4 - 14t^2 - 16t - 3 = 0$

$\Rightarrow (t^2 + 4t + 3)(t^2 - 4t - 1) = 0$

$\Rightarrow t^2 + 4t + 3 = 0$ as $t^2 - 4t - 1 = 0$ can't zero as $(t^2, 2t)$ point can't lies on $x - 2y - 1 = 0$

$\Rightarrow t = -1$ or $t = -3 \Rightarrow$ point are $(1, -2)$ or $(9, -6)$

Section – B

47. 0

Sol. The least value of given expression, $\left(x-(1+\cos t)\right)^2+\left(\frac{1}{2x}-(1+\sin t)\right)^2$ is the shortest distance between two curve $xy = \frac{1}{2}$ and $(x - 1)^2 + (y - 1)^2 = 1$; can be achieved for distance between point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ and $\left(1+\frac{1}{\sqrt{2}}, 1+\frac{1}{\sqrt{2}}\right)$

48. 0

Sol. Let $I = x^{1-z}\int_x^{x+1} \sin t^2 dt$ integration by parts $\int \sin t^2 dt = \frac{-\cos t^2}{2t} - \frac{1}{2}\int \frac{\cos t^2}{t^2} dt$

For $x > 0$, $\int_x^{x+1} \sin t^2 dt = \frac{-\cos(x+1)^2}{2(x+1)} + \frac{\cos(x^2)}{2x} - \frac{1}{2}\int_x^{x+1} \cos t^2 dt$

$\Rightarrow \left|\int_x^{x+1} \sin t^2 dt\right| \le \frac{1}{2(x+1)} + \frac{1}{2x} + \frac{1}{2}\int_x^{x+1} \frac{dt}{t^2} = \frac{1}{x} \Rightarrow \left|x^{1-z}\int_x^{x+1} \sin t^2 dt\right| \le \frac{1}{x^z}$

As $x \to \infty$, $\frac{1}{x^z} \to 0 \Rightarrow \lim_{x\to\infty} x^{-2}\int_x^{x+1} \sin(t^2) dt = 0$

49. 2

Sol. For arbitrary positive x, y, z; $xyz(x + y + z) = 1$

$\Rightarrow (x + y)(y + z) = (x + y + z)y + xz = \frac{1}{xz} + xz \ge 2$ equality hold when $x = z = 1$ and $y = \sqrt{2} - 1$

50. 2

Sol. Let $\left(x + i\sqrt{2}y\right)\left(z + i\sqrt{2}t\right) = 3 + i\sqrt{2}$ by taking modulus

$\Rightarrow \left(x^2 + 2y^2\right)\left(z^2 + 2t^2\right) = 11$

$\Rightarrow \left(x^2 + 2y^2 = 1 \text{ and } z^2 + 2t^2 = 11\right)$ or $\left(x^2 + 2y^2 = 11 \text{ and } z^2 + 2t^2 = 1\right)$

$\Rightarrow (x = 1, y = 0, z = 3, t = 1)$ or $(x = 3, y = 1, z = 1, t = 0)$

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51. 3

Sol. Let consider the polynomial $p(x) = (a_1 + x)(a_2 + x)(a_3 + x) ..... (a_{100} + x) - 1$ as product of all element is each column is 1 the number $b_1, b_2, b_3, ..., b_{100}$ are the root of p(x)

$\Rightarrow p(x) = (x - b_1)(x - b_2)(x - b_3) ..... (x - b_{100})$

$\Rightarrow (x - b_1)(x - b_2)(x - b_3) ..... (x - b_{100}) = (a_1 + x)(a_2 + x)(a_3 + x) ..... (a_{100} + x) - 1$

Now of we substitute $x = -a_i$ into the equation, then $(-1)^{100}(a_i + b_1)(a_i + b_2)(a_i + b_3) ..... (a_i + b_{100}) = -1$

52. 2

Sol. Let assume $f(x) = x^3 - 3x^2 + 5x = (x - 1)^3 + 2(x - 1) + 3$

$\Rightarrow f(\alpha) = 1$ and $f(\beta) = 5$

Now, let $g(y) = y^3 + 2y \Rightarrow g'(y) = 3y^2 + 2 > 0$ monotonic increasing function

$\Rightarrow$ For unique α and $\beta \Rightarrow g(\alpha - 1) = f(\alpha) - 3 = -2$ and $g(\beta - 1) = f(\beta) - 3 = 2$

As f is odd function $\Rightarrow (\alpha - 1) = -(\beta - 1) \Rightarrow \alpha + \beta = 2$

53. 7

Sol. f(x) is increasing function

Max = $f(1) = 8 \Rightarrow a = 2$

Min = $f(-1)$

54. 1

Sol. Let $P\left(x_0, x_0 + \frac{2}{x_0}\right)$. The expression for line PA is $y - \left(x_0 + \frac{2}{x_0}\right) = -\left(x - x_0\right)$

$$\Rightarrow y = -x + 2x_0 + \frac{2}{x_0} \text{ from } \begin{cases} y = x \\ y = -x + 2x_0 + \frac{2}{x_0} \end{cases}$$

We get, $A\left(x_0 + \frac{1}{x_0}, x_0 + \frac{1}{x_0}\right)$ and $B\left(0, x_0 + \frac{2}{x_0}\right)$

$\Rightarrow \overrightarrow{PA} = \frac{1}{x_0}i - \frac{1}{x_0}j$; $\overrightarrow{PB} = -x_0 i$. Therefore, $\overrightarrow{PA} \cdot \overrightarrow{PB} = -1$

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